

ZAID AHMED

AI-ML Engineer | Python Developer

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Block 13/B, Gulshan-e-Iqbal Karachi

SUMMARY

AI/ML Engineer with 3+ years of experience in Python and machine learning, specializing in developing AI solutions for computer vision and NLP. Open-source contributor passionate about advancing AI technology and delivering scalable, high-impact solutions. Recognized for technical expertise, reliability, and driving projects end-to-end from concept to production.

EXPERIENCE

AI-ML Engineer/Python Developer

Inseyab Consulting and Information solutions / Tradeforesight

07/2025 - Present | NASTP, Sharah e Faisal, Karachi

An AI-focused firm building intelligent solutions that enhance business efficiency and strategic decision-making.

- Led end-to-end development of AI-driven solutions, covering backend architecture, model integration, and API design.
- Architected and implemented scalable backend systems ensuring high performance, reliability, and long-term maintainability.
- Designed and developed RESTful APIs using Python frameworks (e.g., FastAPI, Flask) to enable seamless integration between AI models and frontend applications.
- Built and optimized data pipelines for data preprocessing, feature engineering, and efficient model inference workflows.
- Designed robust database schemas and optimized complex queries for relational (PostgreSQL, MySQL) databases.
- Containerized backend services using Docker and managed cloud deployments on AWS and Azure environments.

AI-ML Engineer

Contract Base / Freelance

01/2023 - Present

AI-ML Engineer

Tesseract Innovations (Pvt) Ltd

06/2024 - 06/2025 | Gulshan-e-Iqbal, Karachi

A tech firm focused on innovative AI solutions.

- Designed and deployed end-to-end machine learning models, from data pre-processing to production-ready solutions.
- Applied computer vision techniques for object detection, segmentation, and pose estimation across various use cases.
- Led API integration for seamless model-to-product transitions in real-time applications.
- Performed in-depth data analysis, feature engineering, and hyperparameter tuning to enhance model performance.
- Conducted model evaluation and implemented continuous improvement strategies for deployed systems.
- Implemented Agentic AI concepts to build more autonomous and decision-capable AI systems.

PROJECTS

Voice Intelligence platform

AI-Driven Automated Customer Support System

- Built a real-time conversational AI system with Whisper ASR and natural TTS.
- Enhanced transcription using diarization, normalization, and noise reduction.
- Designed agentic AI workflows with context-aware NLP and smart escalation.
- Implemented vector-based semantic search for accurate information retrieval.
- Utilized Apache Kafka for scalable, low-latency streaming and queue management.

Web3 Trading Chatbot

A crypto trading guide with action-ready agents inspired by Chatoshi platform

- AI-powered crypto assistant designed to guide users in Web3 trading.
- Understands user commands and executes trading actions like: Buy / Sell / Transfer crypto & Copy trading from other wallets or traders
- Action-ready agents can provide real-time market updates, trade execution, and strategy suggestions.
- Uses natural language input to simplify complex crypto trading tasks.
- Voice command support for hands-free interaction
- Built as an intelligent crypto guide + trading assistant. Ideal for both beginners and experienced traders navigating decentralized markets.

Twitter/X Engagement Agent

A custom-built Twitter/X bot for the official Game5Ball page.

- Automated engagement: Interacts with followers through replies, likes, and retweets to boost visibility.
- Content generation: Capable of posting dynamic, AI-generated tweets to drive engagement.
- Growth-oriented: Designed to enhance community interaction and brand presence.
- Scalable features: Future updates planned to include deeper analytics, trend monitoring, and campaign automation.

📁 EXPERIENCE

Python Developer (Internship)

Codes Soft

📅 10/2023 - 12/2023

📍 Gulshan-e-Iqbal, Karachi

A technology company specializing in ERP software development.

Contributed to feature development and customized ERP applications.

Collaborated on real-world projects and enhanced Python development skills.

🎓 EDUCATION

Bachelors of Science in Software Engineering

Sir Syed University of Engineering & Technology

📅 08/2020 - 06/2024

📍 Karachi

Artificial Intelligence and Data Sciences

Saylani Mass IT (SMIT)

📅 01/2022 - 12/2023

📍 Karachi

⚙️ SKILLS

AI EXPERTISE

Python

Generative AI (GenAI)

Agentic AI

RAG

Computer Vision

OCR

Text To Speech

Tensorflow

PyTorch

Feature Engineering

Data Pre-processing

Data Annotation

Image Analysis

Video Analysis

Voice Analysis

Vector Databases

Scikit-learn

Fine-tuning Models

Model Deployment

LangChain

LangGraph

ENGINEERING & INFRASTRUCTURE

Backend Development (FastAPI/Django/Flask)

NVIDIA Triton

Roboflow

CI/CD

Claude Code/API

OpenAI API

Docker

Kafka

Redis

PostgreSQL

MySQL

CLOUD PLATFORMS & TOOLS

AWS

Azure

GCP

Vast.ai

RunPod

⚙️ PROJECTS

MeasureMates

An AI project focused on measuring and analyzing cow postures.

Developed a 2D pose estimation & segmentation model of livestock animals (cows).

Managed every step from data collection and annotating process to model training and deployment.

Backend development for seamless integration via API's

📄 PUBLICATIONS

Open Source Datasets

Kaggle

📅 04/2024

kaggle.com/datasets/zaidworks0508

Led the gathering, labelling, analysis and model training process.

🏆 CERTIFICATIONS

- ✓ AI and Data Science From SMIT

✓ Building With Claude API By Anthropic

✓ Introduction To Model Context Protocol By Anthropic

🏆 AWARDS

- 🏆 Sir Syed Technovative Ideathon Winner

Winner of Sir Syed Technovative Ideathon 2024.

🏆 Devathon SUMIT 1.0 Laptop Winner

Won the Laptop in 'AI & DS' Category at Devathon SUMIT 1.0.

🏆 2nd Position in AI Project Exhibition

Achieved 2nd Position in University AI Project Exhibition 2022.

🗣️ LANGUAGES

English

Proficient

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Urdu

Native

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